Beyond the Cheapest Fleet: Safety-Driven Optimization

A Cost-Safety Trade-off in Monthly Bunker Transport

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The Fleet Selection Problem

1. Moves monthly bunker cargo
2. Minimizes total system cost
3. Meets safety and fuel diversity
4. Uses each vessel only once

Core Challenge: Balancing cost, safety, and emission in one decision

Two Safety Scenarios

Baseline Fleet

- Average Safety ≥ 3
- Cost optimal under standard conditions

High-safety fleet

- Average Safety ≥ 4
- Represents stricter industry expectations

Baseline	High-Safety
Safety ≥3	Safety ≥4
Cost focus	Safety focus

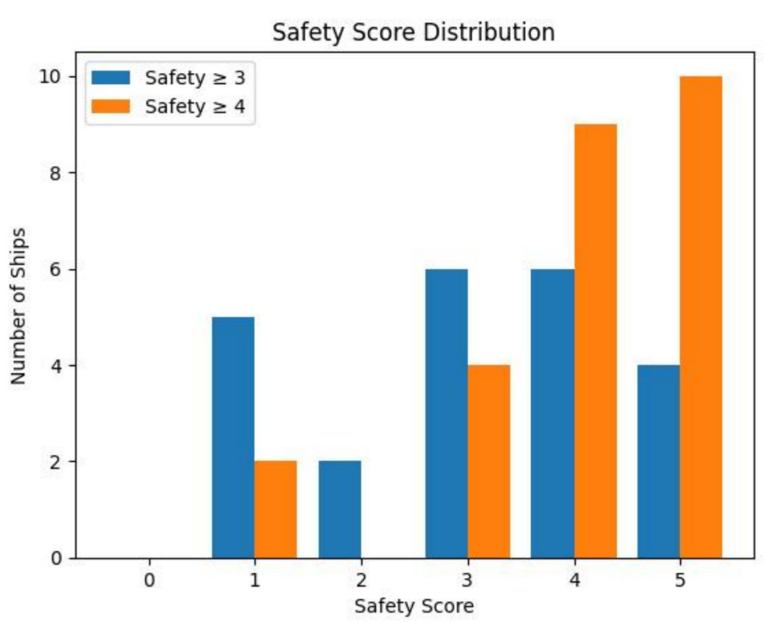
Goal: Understand how higher safety changes cost and fleet structure

Baseline vs High-Safety Fleet

Metric	Safety ≥ 3	Safety ≥ 4
Fleet size	23 ships	25 ships
Total cost	$1.53B	$1.69B
Avg safety	3.09	4.00
Emissions	56k t CO_2e	52k t CO_2e

Key takeaway: +10% cost → significantly safer fleet with lower emissions

What Changes When Safety Increases?



− Low-safety vessels

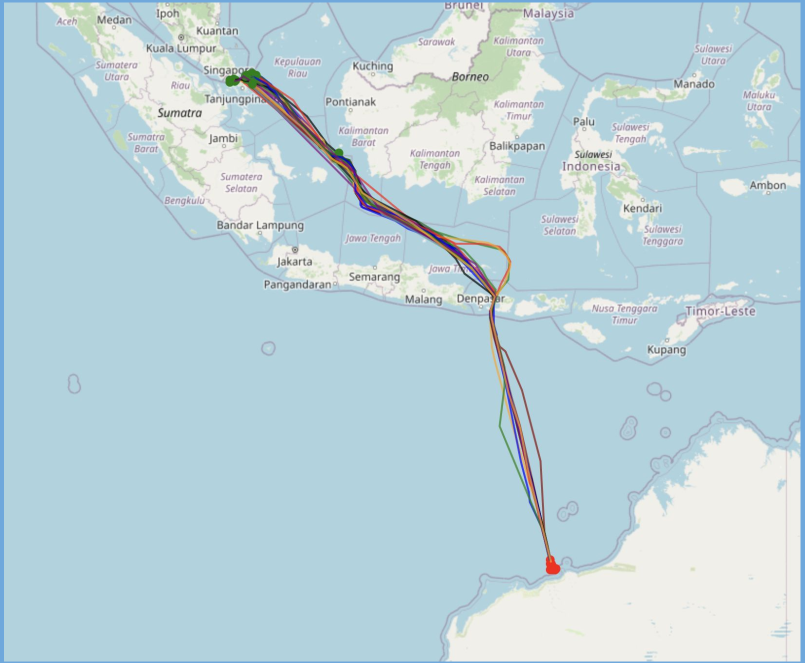
+ High-safety vessels

+ Fleet size

Insight: Safety acts as a binding constraint in the model

Operational Realism and Final Takeaways

Selected vessels follow real AIS-derived routes between Singapore and Australia



Final takeaway:
Higher safety standards raise costs, but also reduce emissions and operational risk